

Grid Features Validation Packet -- MXIC4S

Every trace in this packet is simulated output from a fitted, multi-conductance single-compartment model of this cell, integrated deterministically -- not a raw current-clamp recording. What is being evaluated here is the feature-extraction methodology applied on top of that model: spike detection, burst/tonic classification, post-inhibitory rebound detection, sag depth, and spike-frequency adaptation. Each figure below reruns the actual classification algorithm on a real simulated trace and marks what it found directly on the trace.

Silencing threshold: -4.75 nA | 2500 points sampled across the current grid

Burstiness index: 0.630

Firing-rate/current slope: 5.72 Hz/nA ($R^2=0.975$)

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Every extracted-feature panel across the full current grid -- what the rest of this packet verifies.

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Every distinct response pattern this cell produces, then one full-detail trace per pattern.

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Every spike the automated detector found, marked on real traces.

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The baseline voltage, search window, and trough behind the sag measurement.

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Early vs. late interspike intervals on a real, sustained tonic trace.

12. Firing rate vs. injected current

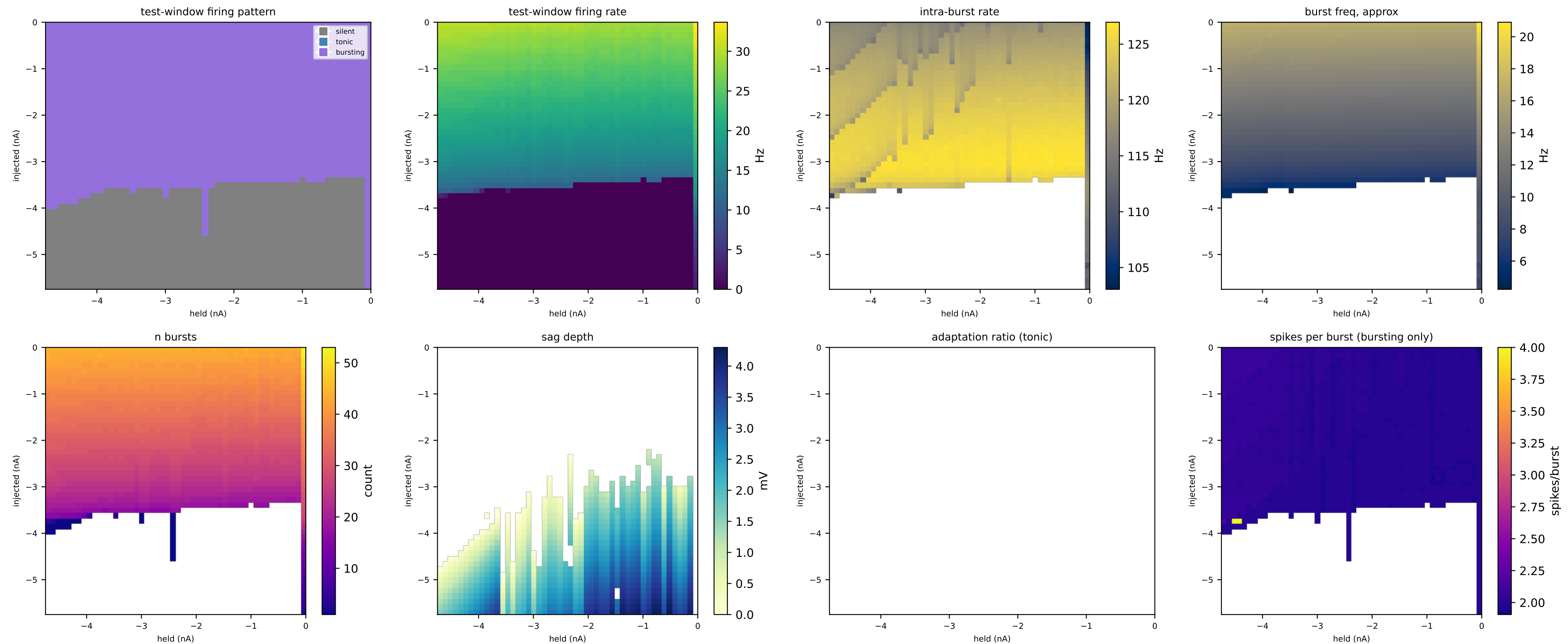
Firing rate vs. current, with the linear fit used to summarize it.

13. Reproducibility check

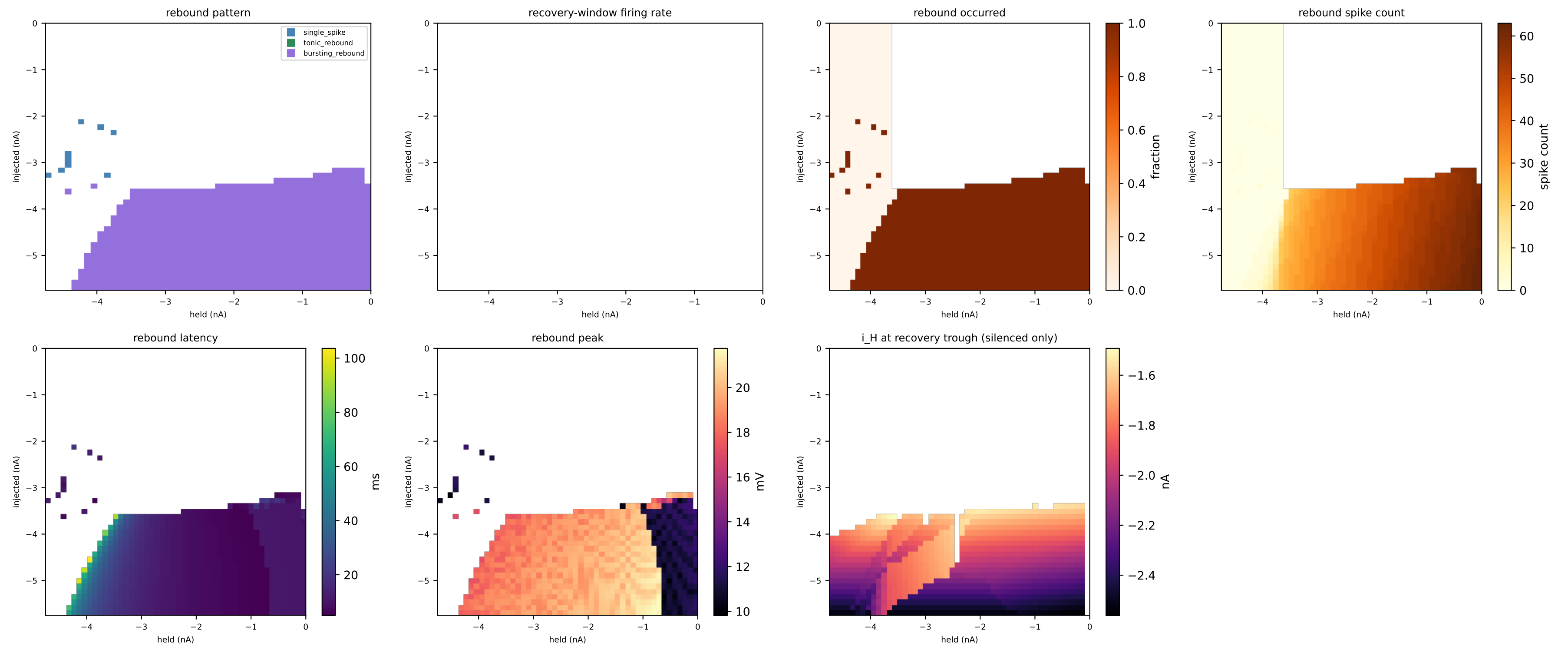
How often re-simulation reproduces the original classification.

1. Feature overview: MXIC4S's full current grid

TEST PHASE



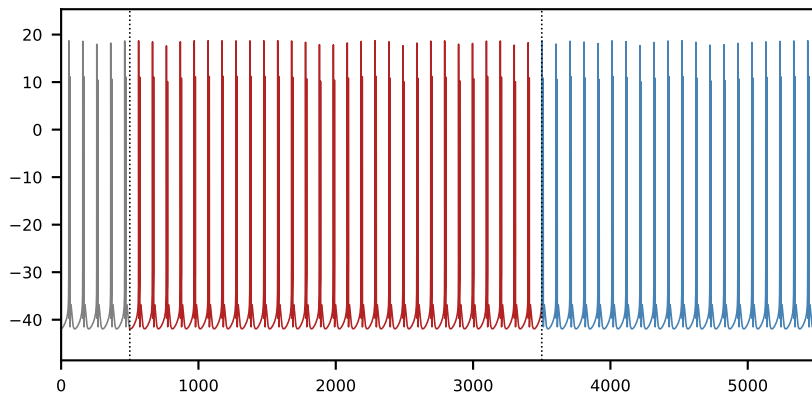
RECOVERY PHASE



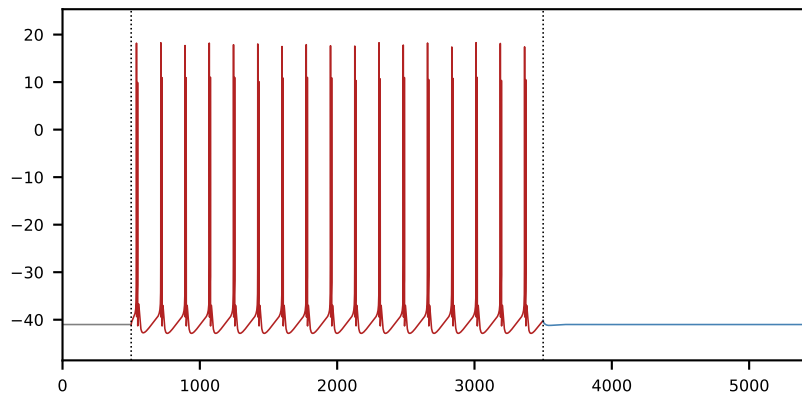
Every panel below is computed from the same 2500-point grid of held and injected current levels: which pattern each point fires in and whether it rebounds after release from hyperpolarization (top row), firing rate and burst structure (second row), rebound spike count, latency, and peak voltage (third row), and sag depth, adaptation, and burst size (bottom row). Two single-number summaries condense the whole grid: a burstiness index of 0.630 (the fraction of points classified bursting) and a firing-rate-vs-current slope of 5.72 Hz/nA ($R^2=0.975$). The rest of this packet does not add new results -- it re-derives individual pieces of this same grid from real simulated traces so each one can be checked by eye against the panel it feeds.

2. Representative traces

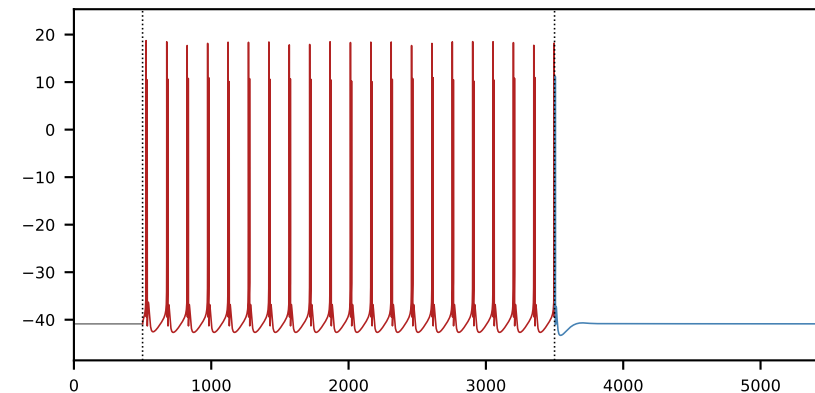
bursting, then a burst-like rebound
held=-2.13 nA, injected=-2.11 nA



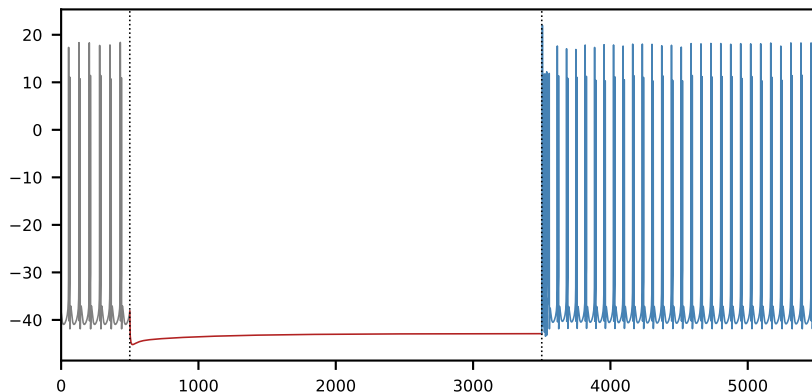
bursting, then no rebound
held=-3.97 nA, injected=-3.52 nA



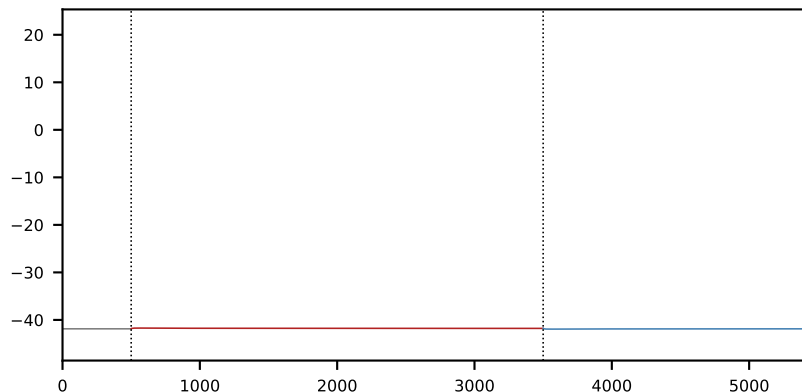
bursting, then a single-spike rebound
held=-3.88 nA, injected=-3.29 nA



silent (no firing), then a burst-like rebound
held=-0.48 nA, injected=-5.63 nA

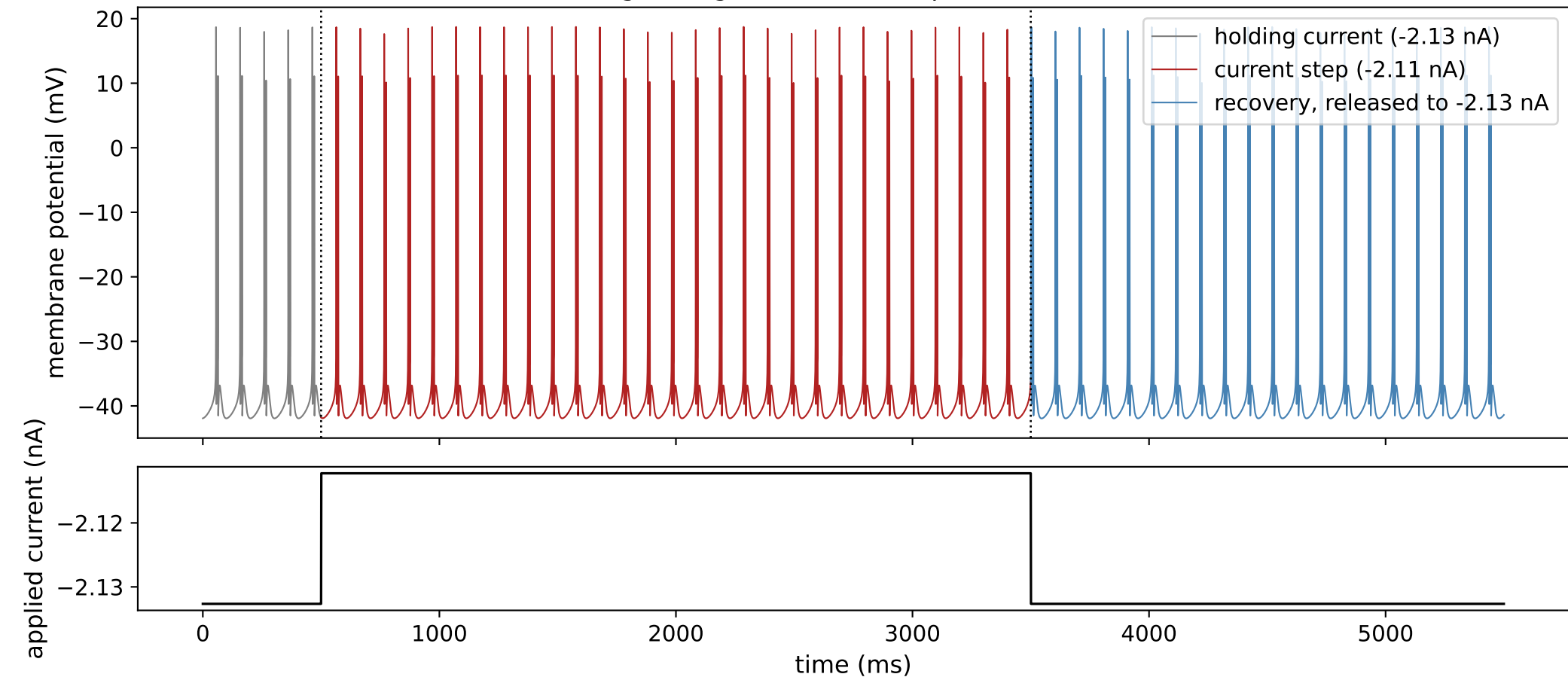


silent (no firing), then no rebound
held=-4.65 nA, injected=-4.58 nA

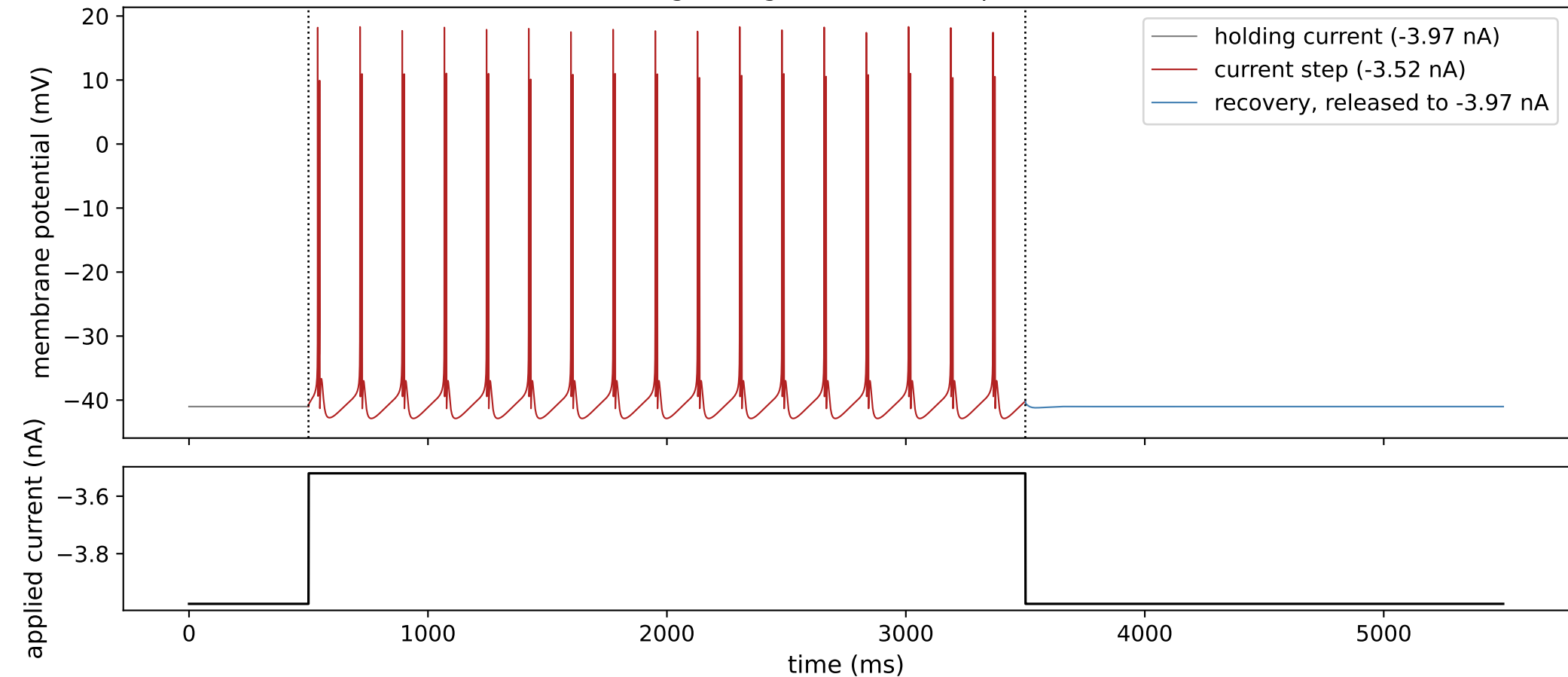


One representative trace for each distinct combination of response-to-current-step pattern and rebound pattern observed across this cell's full 2500-point grid of held and injected current levels: gray is the held baseline before the current step, red is the response during the current step, and blue is the recovery period after release back to the held level. Each combination is represented by its most clearly-expressed example among the grid points that produced it. Every panel shares the same voltage (-49 to 25 mV) and time (0-5500 ms) axes so amplitude and duration are directly comparable by eye -- without this, a small subthreshold wobble and a full-amplitude spike train would each be independently rescaled to fill the same

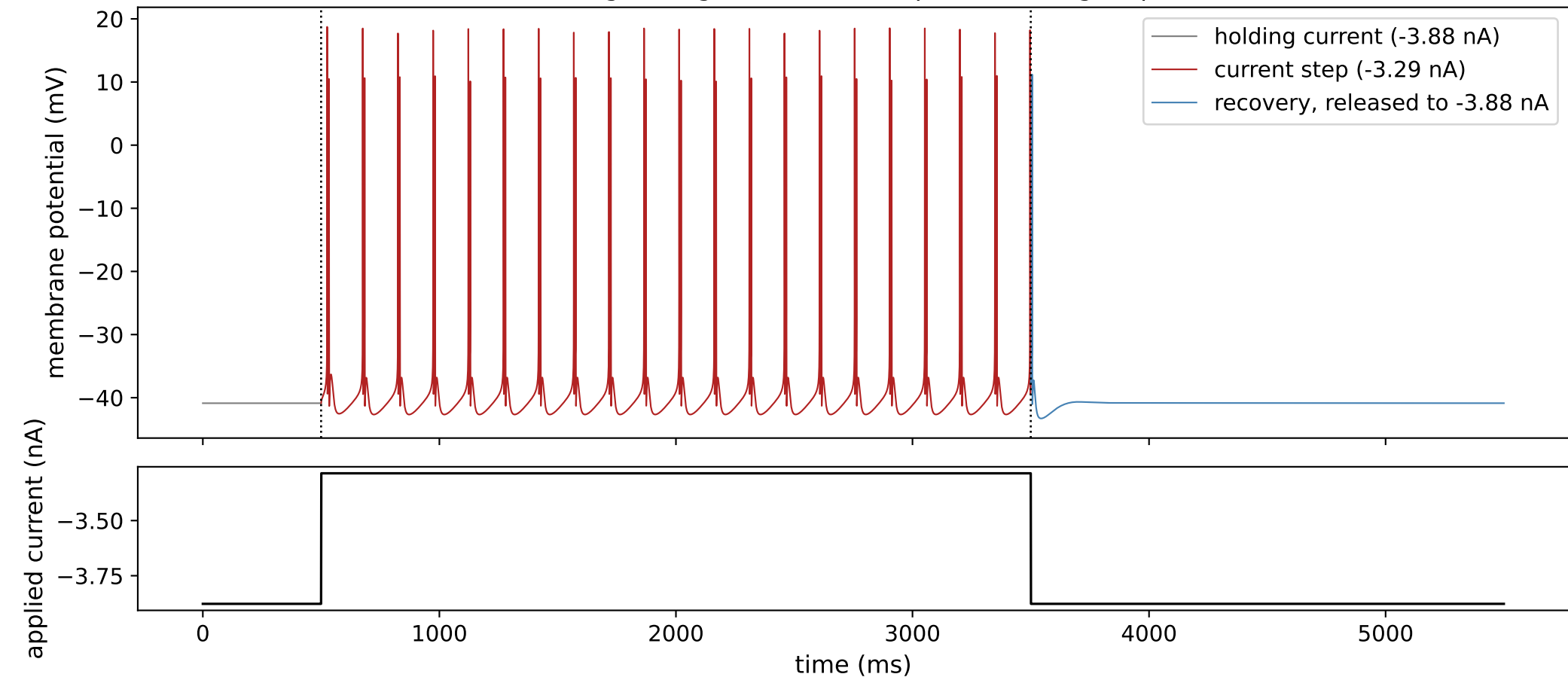
MXIC4S: bursting during the current step, then a burst-like rebound



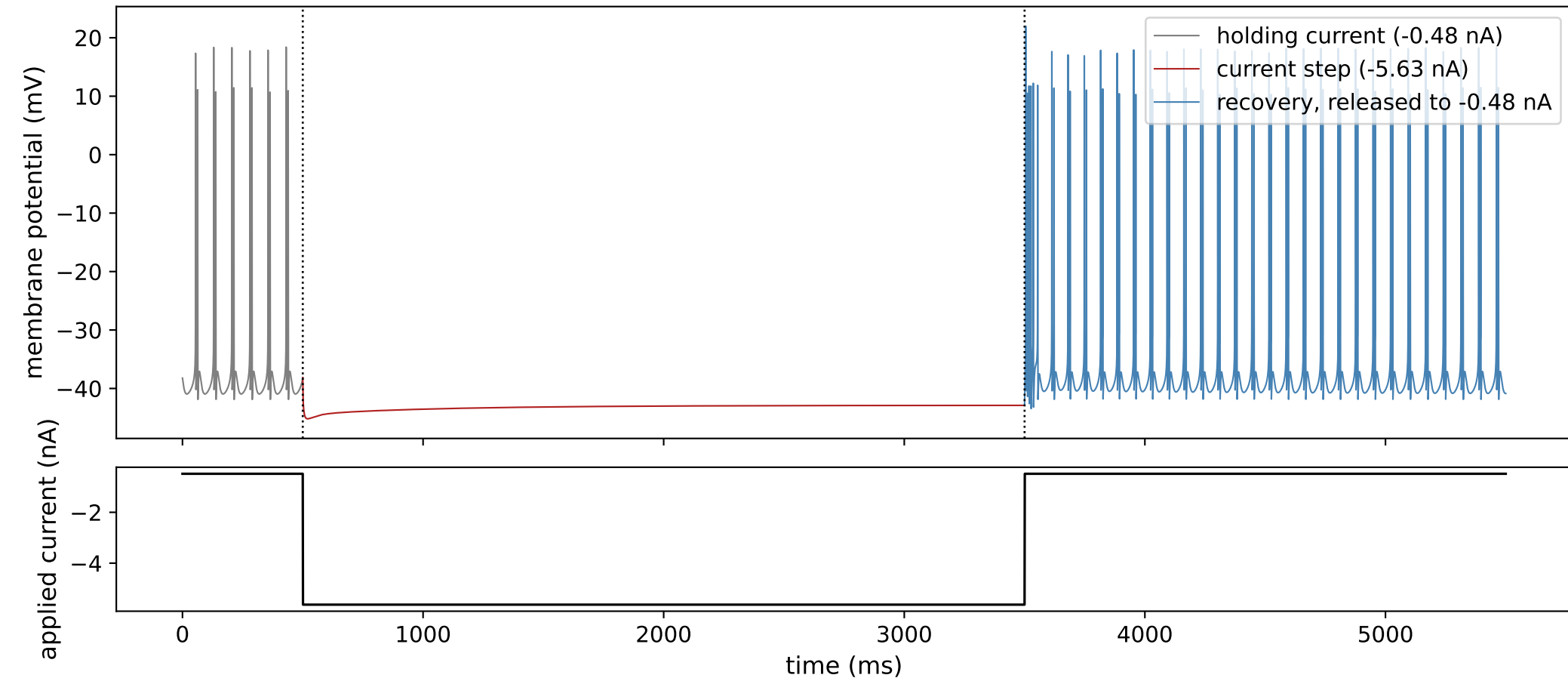
MXIC4S: bursting during the current step, then no rebound



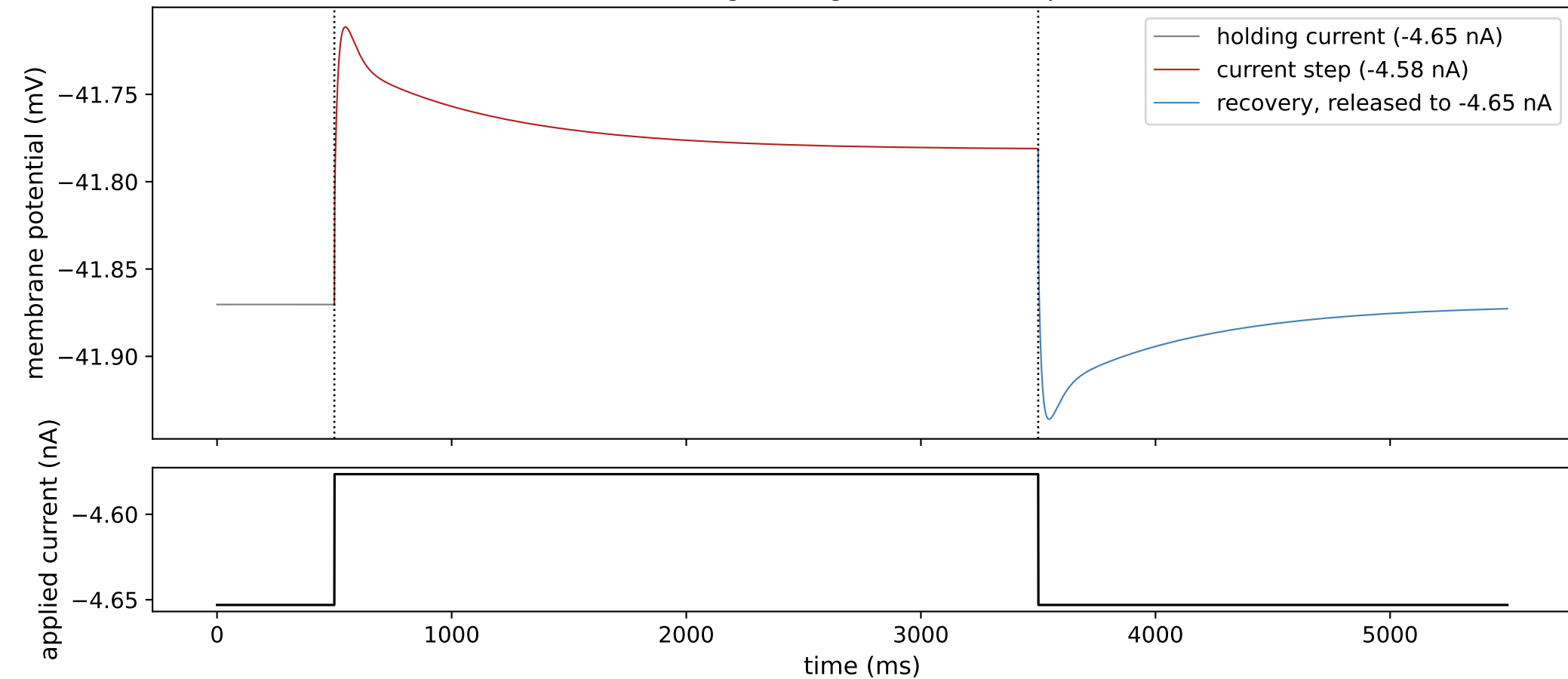
MXIC4S: bursting during the current step, then a single-spike rebound



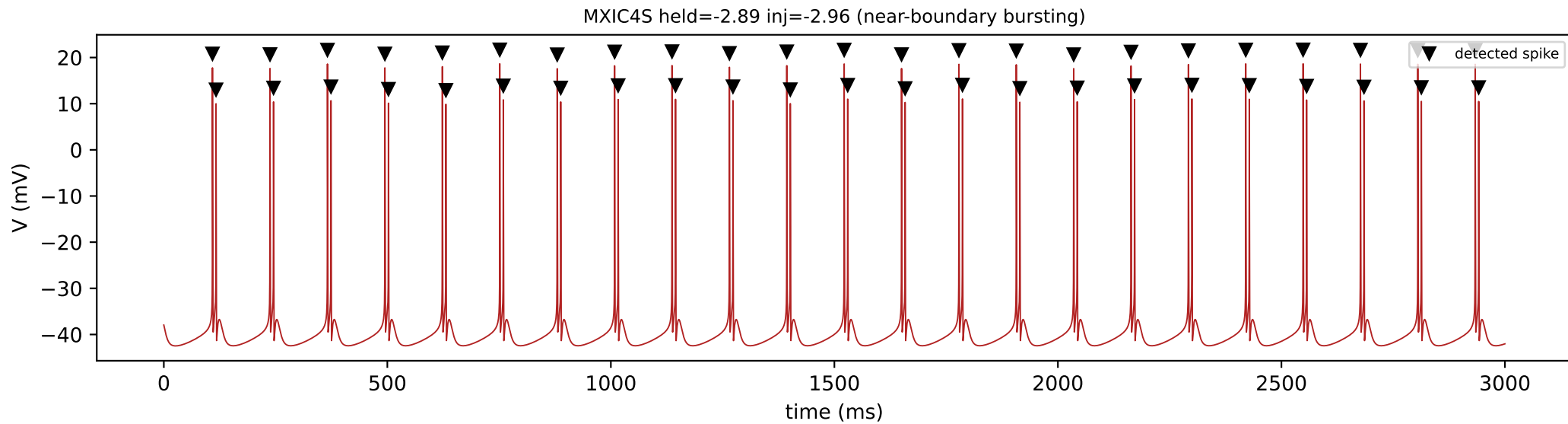
MXIC4S: silent (no firing) during the current step, then a burst-like rebound



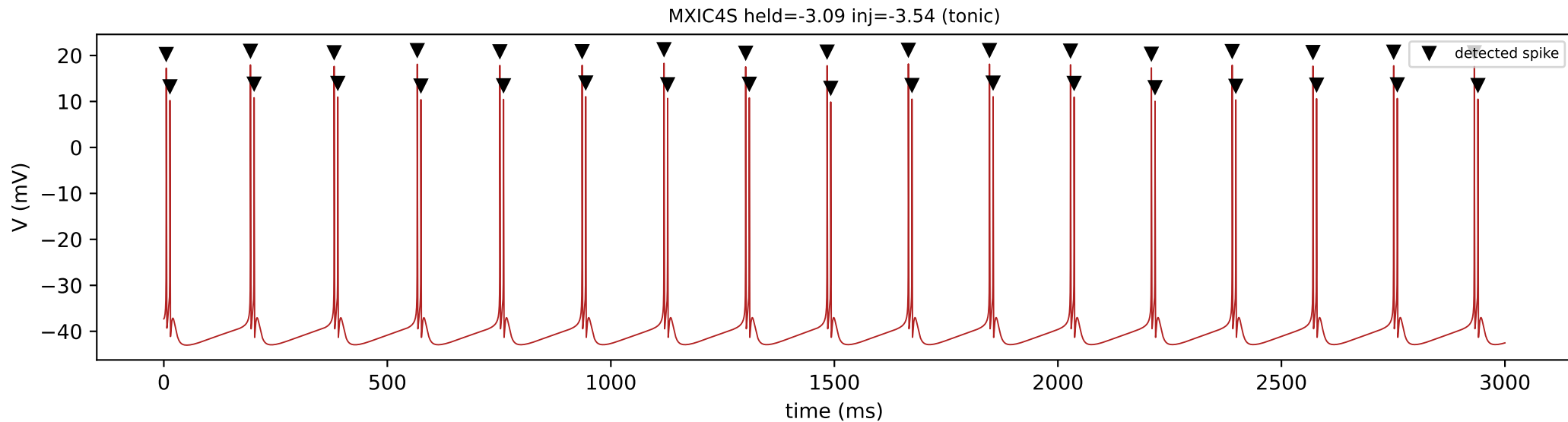
MXIC4S: silent (no firing) during the current step, then no rebound



3. Spike detection

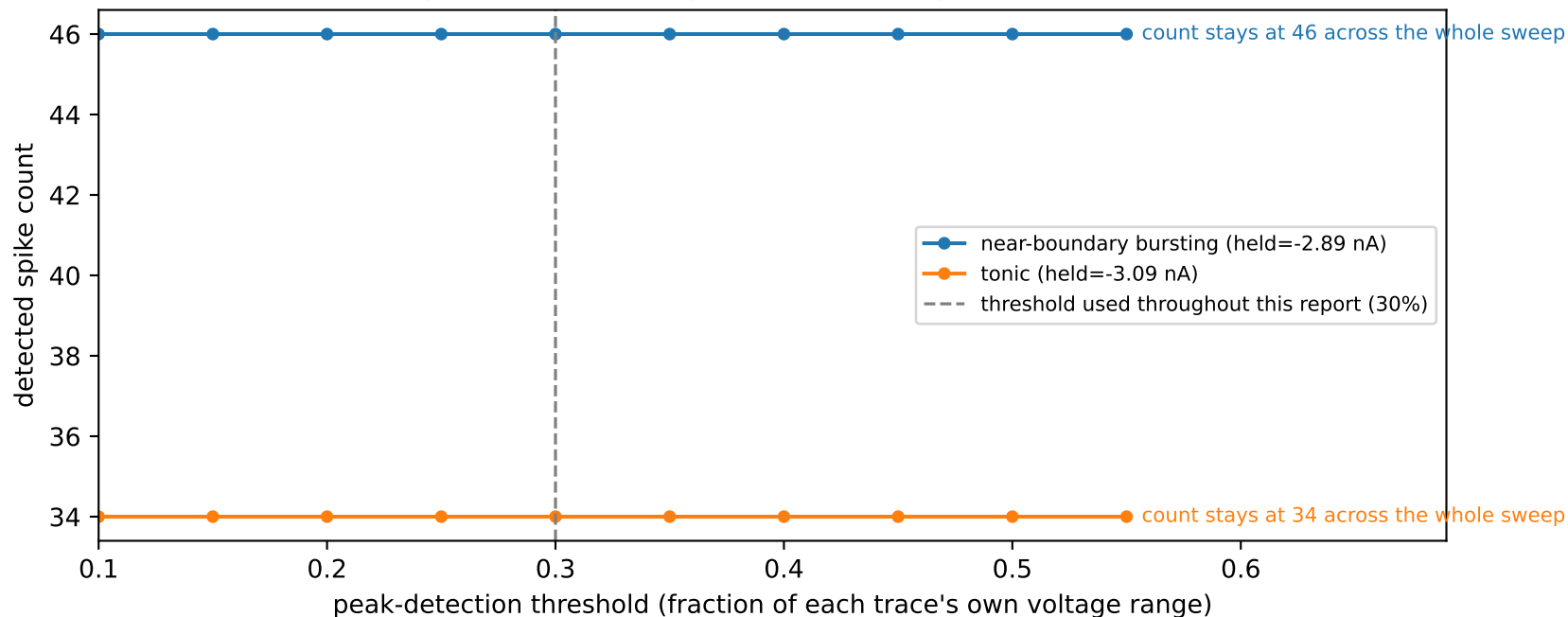


Markers show every spike this trace's automated peak detector found, requiring each peak to rise at least 18.3 mV above its surroundings (30% of this trace's own 61.1 mV voltage range) -- this is the same spike count used for every interspike-interval and burst/tonic classification elsewhere in this report. 46 spike(s) were detected in this window.



4. Spike-detection threshold sensitivity

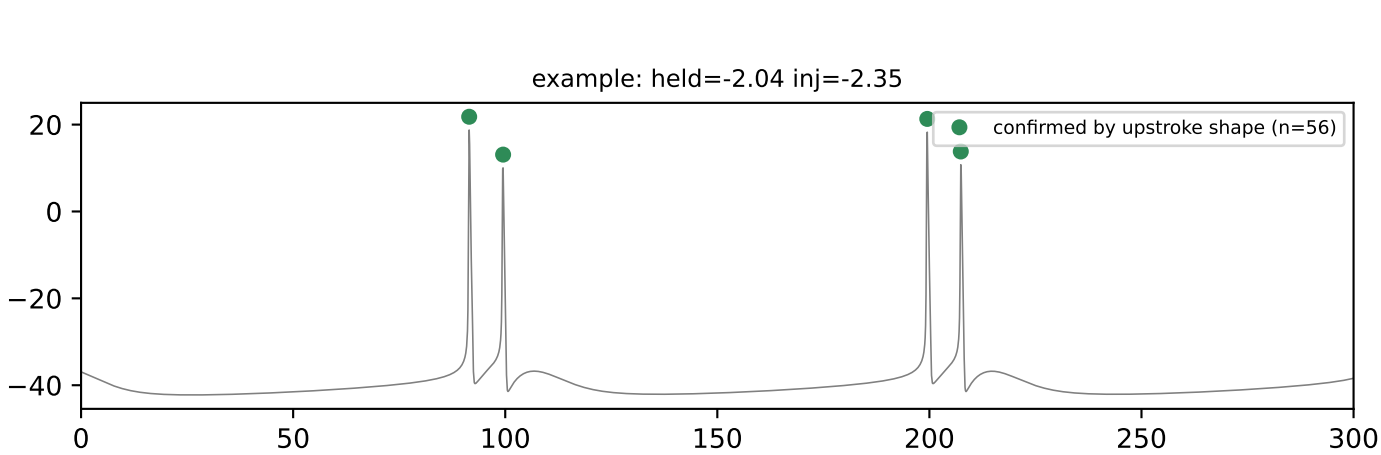
MXIC4S -- spike count is unchanged across a 5x range of detection thresholds



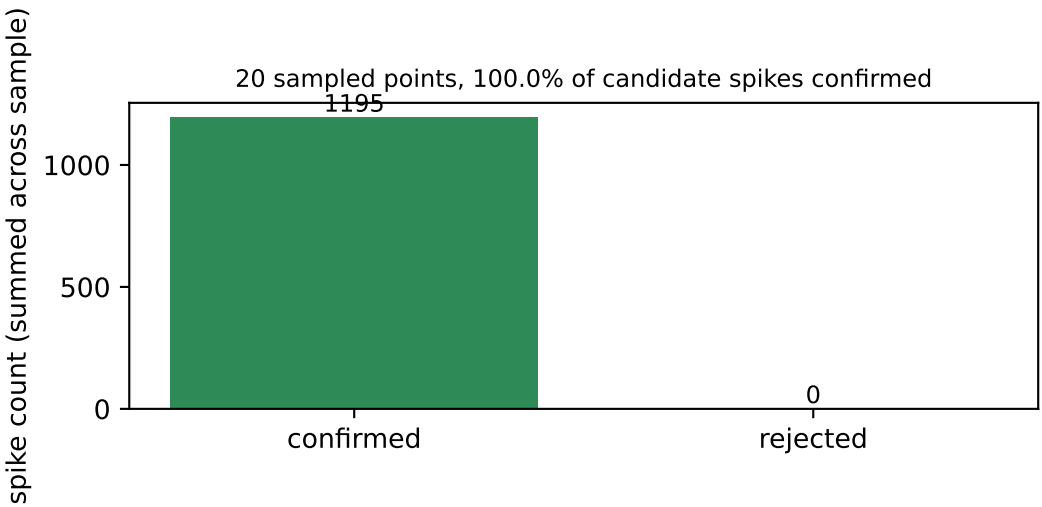
Every spike count, interspike interval, and burst/tonic classification in this report depends on one detection threshold: a candidate peak must rise at least 30% of the trace's own voltage range above its surroundings (or 15 mV, whichever is larger, so a tiny subthreshold wobble in a low-amplitude trace cannot be counted as a spike). This sweeps that threshold from 10% to 55% on real, large-amplitude spiking traces: a flat plateau around the value used throughout this report shows the reported spike counts are insensitive to the exact threshold chosen.

```
python src/generate_validation_packet.py --cells XB2IQX DFJ1K3 8Z3ESD MXIC4S
source: cell_held_injected_grid.pkl -> resimulate_point() -> scipy.signal.find_peaks swept over PROMINENCE_FRACTION
```

5. Independent cross-check of spike detection by upstroke shape

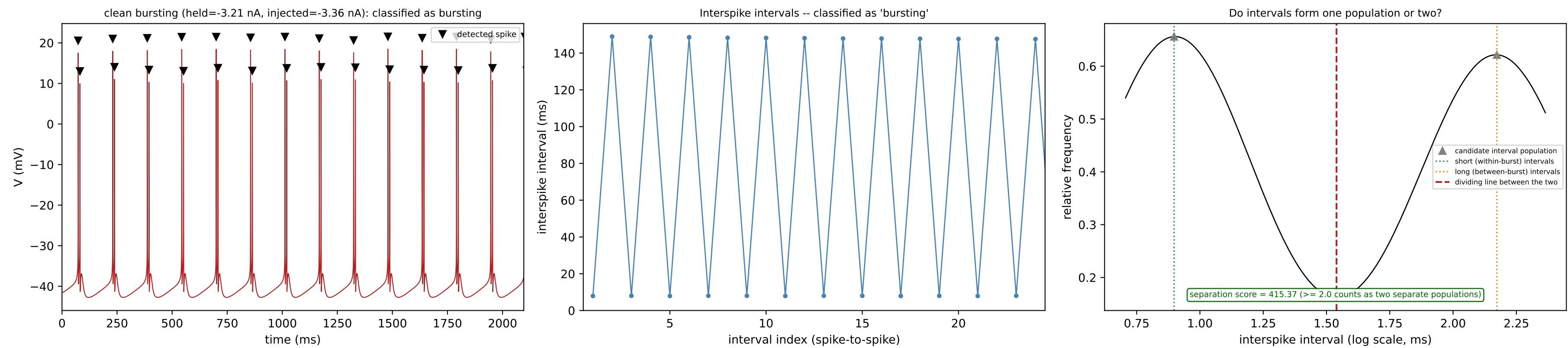


As an independent check, each candidate spike was also required to show a fast upstroke -- a rate of voltage rise of at least 10 mV/ms within 20 ms before the peak (the convention used by Bean, 2007) -- rather than amplitude alone. 56 of 56 candidate spike(s) were confirmed by this stricter test; the two methods agree on every spike in this window.

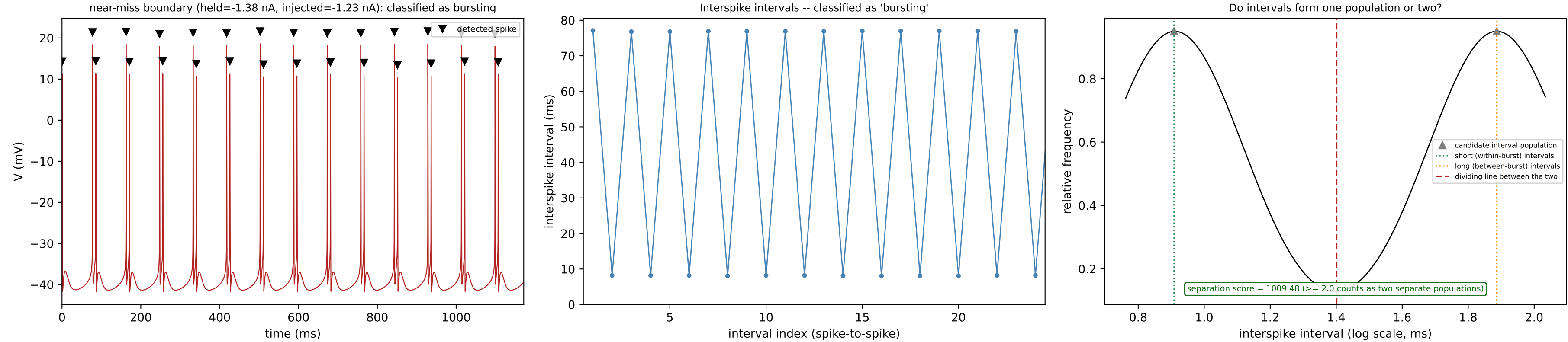


As an independent check on a random sample of 20 grid points from this cell, every spike detected by amplitude alone was additionally required to show a fast upstroke in voltage (rather than amplitude alone) before being counted, an independent detection criterion not used for classification elsewhere in this report. Agreement close to 100% shows that the simpler amplitude-based detector used throughout is not missing any spikes that a shape-based check would catch.

6. Tonic / bursting / silent classification

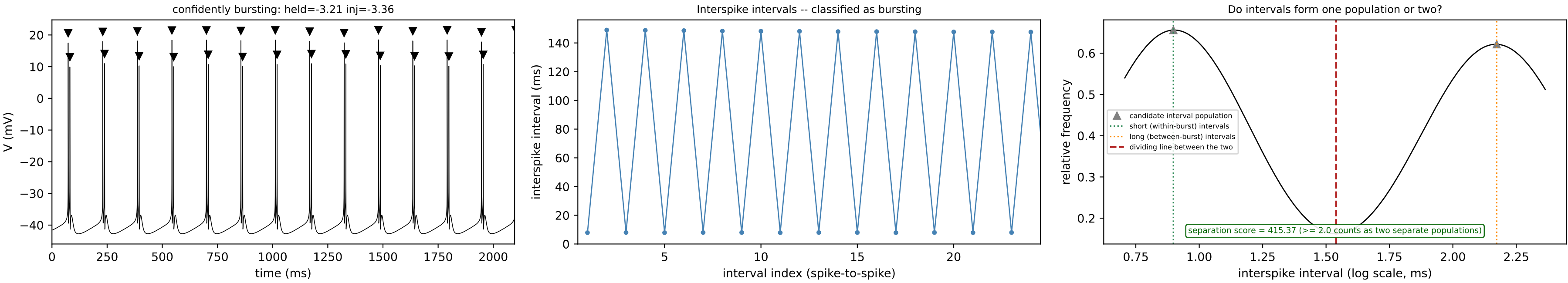


This window contained 38 spikes, giving 37 interspike intervals. Splitting the intervals at their 2 candidate population(s) gives a short (within-burst) interval of 8.0 ms and a long (between-burst) interval of 147.9 ms -- a 18.58-fold difference, above the 1.5-fold difference required to treat them as distinct. On average, 2.00 spikes fall within each burst (a real burst requires at least 1.5, ruling out what would otherwise just be occasional missed spikes in an otherwise tonic train). A statistical separation score (Ashman's D) of 415.37 quantifies how cleanly the short and long intervals separate into two distinct populations (a score of 2.0 or higher is required to call them separate). Taken together, this window is classified as 'bursting', with a separation score of 415.37.

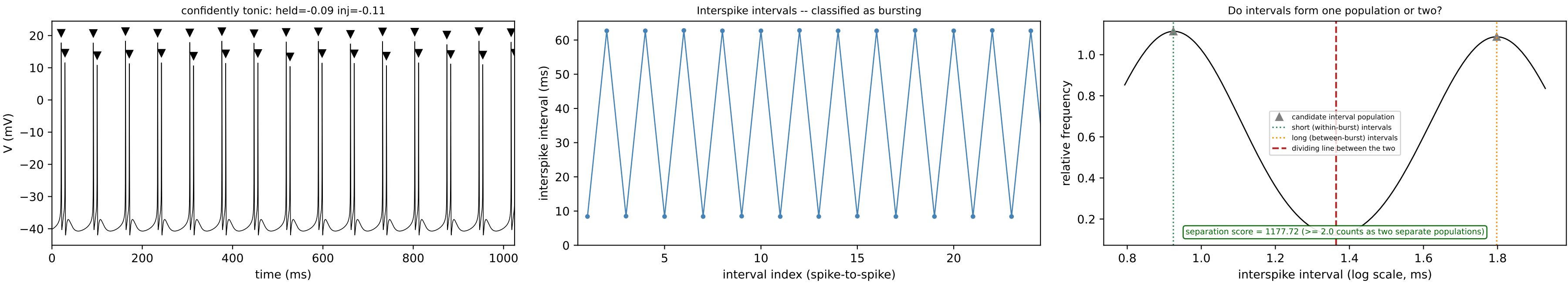


This window contained 71 spikes, giving 70 interspike intervals. Splitting the intervals at their 2 candidate population(s) gives a short (within-burst) interval of 8.1 ms and a long (between-burst) interval of 77.0 ms -- a 9.45-fold difference, above the 1.5-fold difference required to treat them as distinct. On average, 1.97 spikes fall within each burst (a real burst requires at least 1.5, ruling out what would otherwise just be occasional missed spikes in an otherwise tonic train). A statistical separation score (Ashman's D) of 1009.48 quantifies how cleanly the short and long intervals separate into two distinct populations (a score of 2.0 or higher is required to call them separate). Taken together, this window is classified as 'bursting', with a separation score of 1009.48.

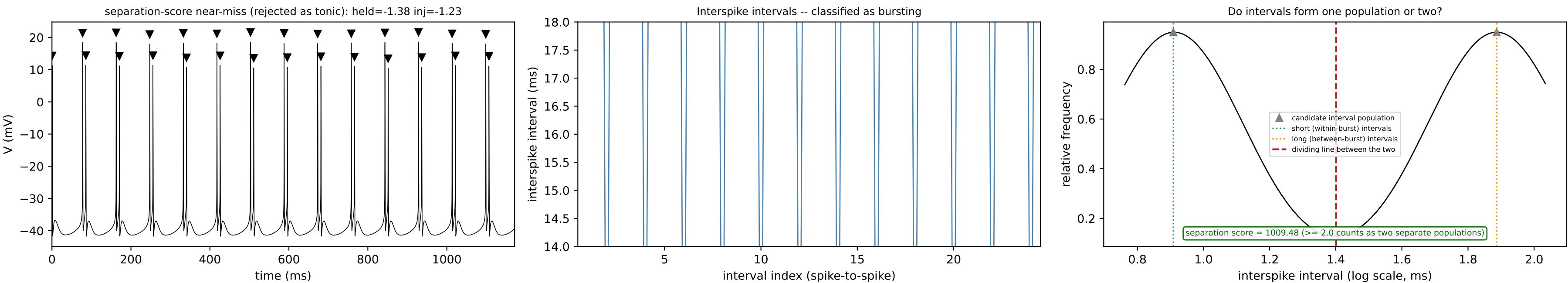
7. Bimodality test: within-burst vs. between-burst ISIs



Simulated MXIC4S trace. This window contained 38 spikes, giving 37 interspike intervals. Splitting the intervals at their 2 candidate population(s) gives a short (within-burst) interval of 8.0 ms and a long (between-burst) interval of 147.9 ms -- a 18.58-fold difference, above the 1.5-fold difference required to treat them as distinct. On average, 2.00 spikes fall within each burst (a real burst requires at least 1.5, ruling out what would otherwise just be occasional missed spikes in an otherwise tonic train). A statistical separation score (Ashman's D) of 415.37 quantifies how cleanly the short and long intervals separate into two distinct populations (a score of 2.0 or higher is required to call them separate). Taken together, this window is classified as 'bursting', with a separation score of 415.37.

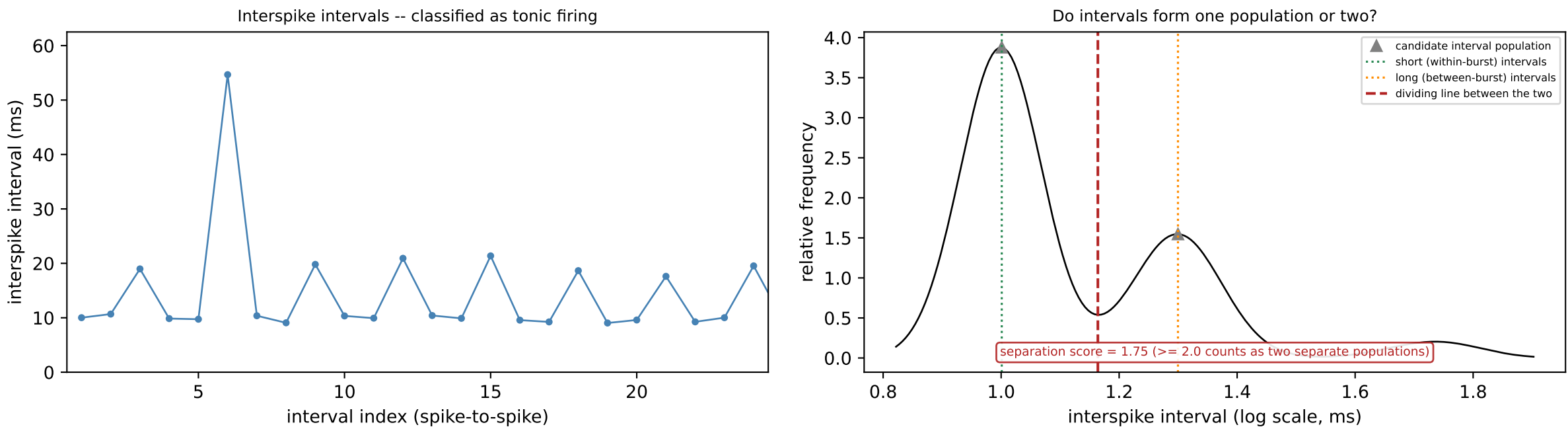


Simulated MXIC4S trace. This window contained 84 spikes, giving 83 interspike intervals. Splitting the intervals at their 2 candidate population(s) gives a short (within-burst) interval of 8.4 ms and a long (between-burst) interval of 62.7 ms -- a 7.45-fold difference, above the 1.5-fold difference required to treat them as distinct. On average, 2.00 spikes fall within each burst (a real burst requires at least 1.5, ruling out what would otherwise just be occasional missed spikes in an otherwise tonic train). A statistical separation score (Ashman's D) of 1177.72 quantifies how cleanly the short and long intervals separate into two distinct populations (a score of 2.0 or higher is required to call them separate). Taken together, this window is classified as 'bursting', with a separation score of 1177.72.



Simulated MXIC4S trace. This window contained 71 spikes, giving 70 interspike intervals. Splitting the intervals at their 2 candidate population(s) gives a short (within-burst) interval of 8.1 ms and a long (between-burst) interval of 77.0 ms -- a 9.45-fold difference, above the 1.5-fold difference required to treat them as distinct. On average, 1.97 spikes fall within each burst (a real burst requires at least 1.5, ruling out what would otherwise just be occasional missed spikes in an otherwise tonic train). A statistical separation score (Ashman's D) of 1009.48 quantifies how cleanly the short and long intervals separate into two distinct populations (a score of 2.0 or higher is required to call them separate). Taken together, this window is classified as 'bursting', with a separation score of 1009.48.

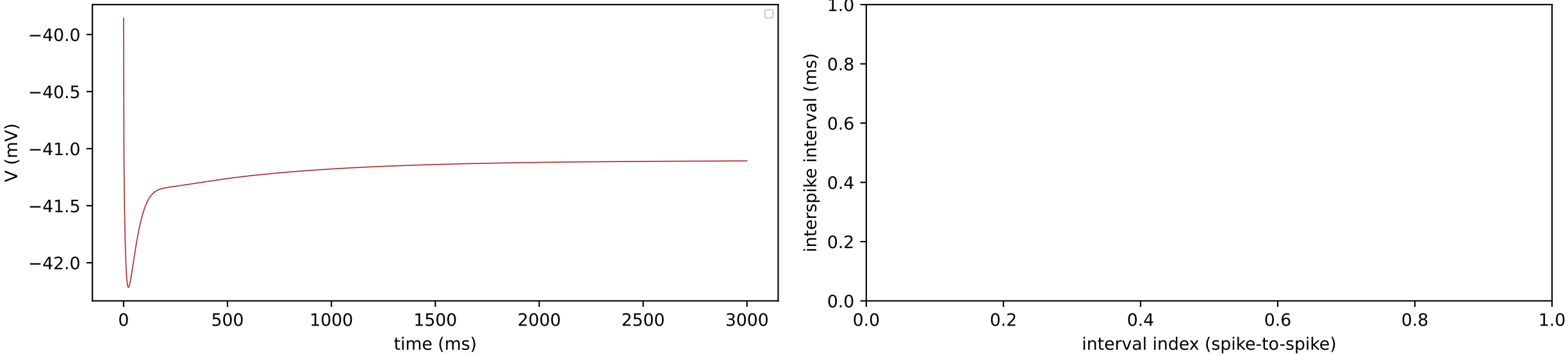
Constructed example
(a hand-built interval sequence,
not a simulated trace)



This example is constructed, not an observed point from MXIC4S: it illustrates a case where the short and long interval populations pass the simpler gates but remain too statistically overlapping to count as separate, a case this cell's own data has not been observed to produce. This window contained 180 spikes, giving 179 interspike intervals. Splitting the intervals at their 2 candidate population(s) gives a short (within-burst) interval of 10.0 ms and a long (between-burst) interval of 24.0 ms -- a 2.40-fold difference, above the 1.5-fold difference required to treat them as distinct. On average, 3.00 spikes fall within each burst (a real burst requires at least 1.5, ruling out what would otherwise just be occasional missed spikes in an otherwise tonic train). A statistical separation score (Ashman's D) of 1.75 quantifies how cleanly the short and long intervals separate into two distinct populations (a score of 2.0 or higher is required to call them separate). Taken together, this window is classified as 'tonic', with a separation score of 1.75.

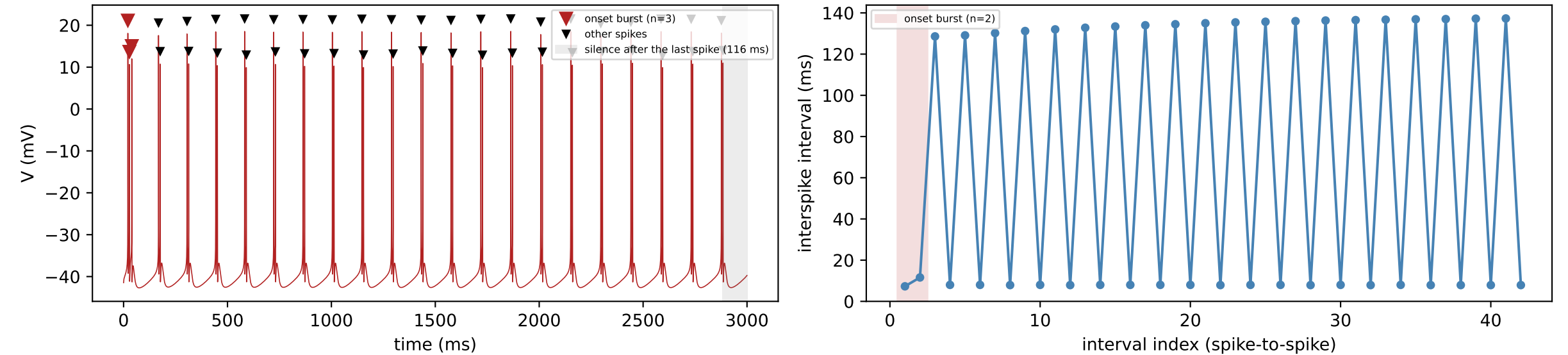
8. Burst-onset-then-silence detection

burst then ceased (held=-2.48 nA, injected=-4.04 nA): classified as silent (no firing), adaptation ratio not reported (see caption)



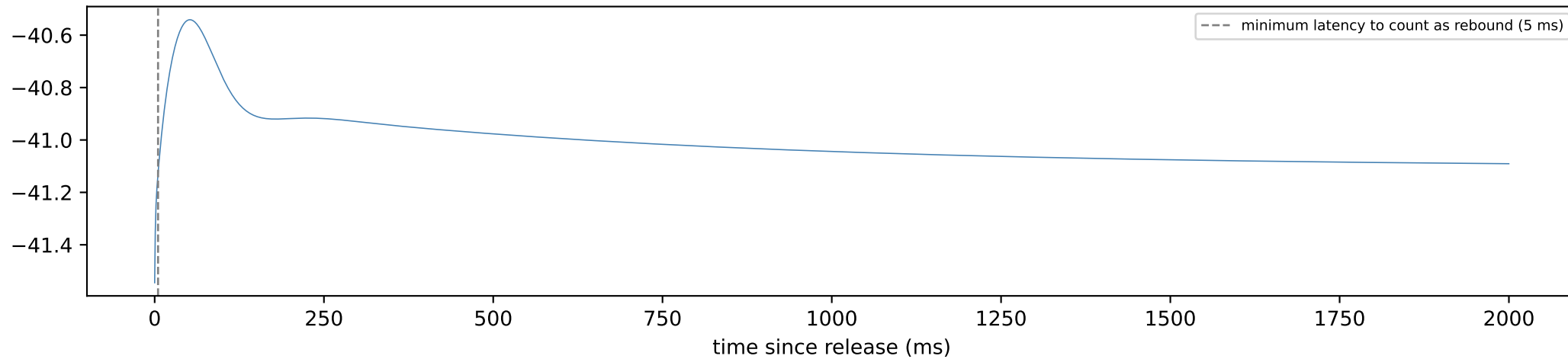
No burst-shaped leading run of spikes occurred at the start of this window.

burst then sustained (held=-4.32 nA, injected=-3.26 nA): classified as bursting, adaptation ratio not reported (see caption)



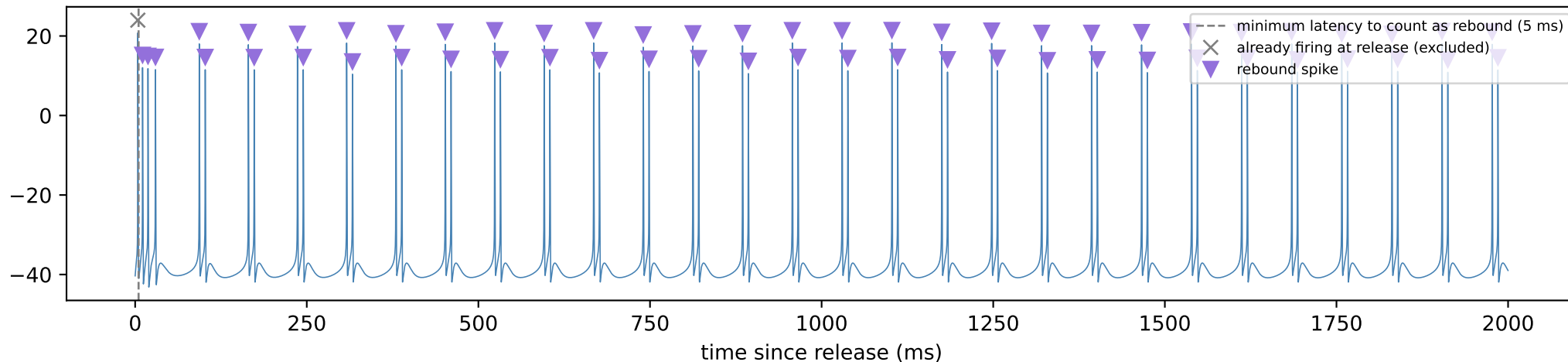
9. Post-inhibitory rebound detection

single-spike rebound: held=-4.04 inj=-4.38



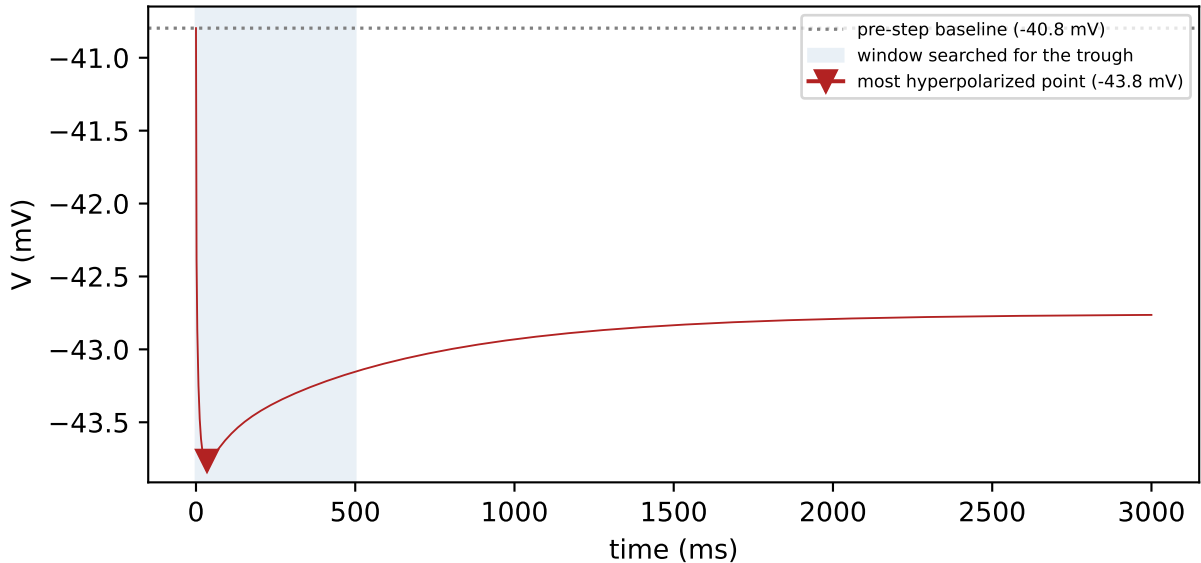
0 spike(s) occurred at or after 5 ms post-release and so count as post-inhibitory rebound. This cell did not show a rebound response.

sustained (tonic) rebound: held=-0.32 inj=-3.54



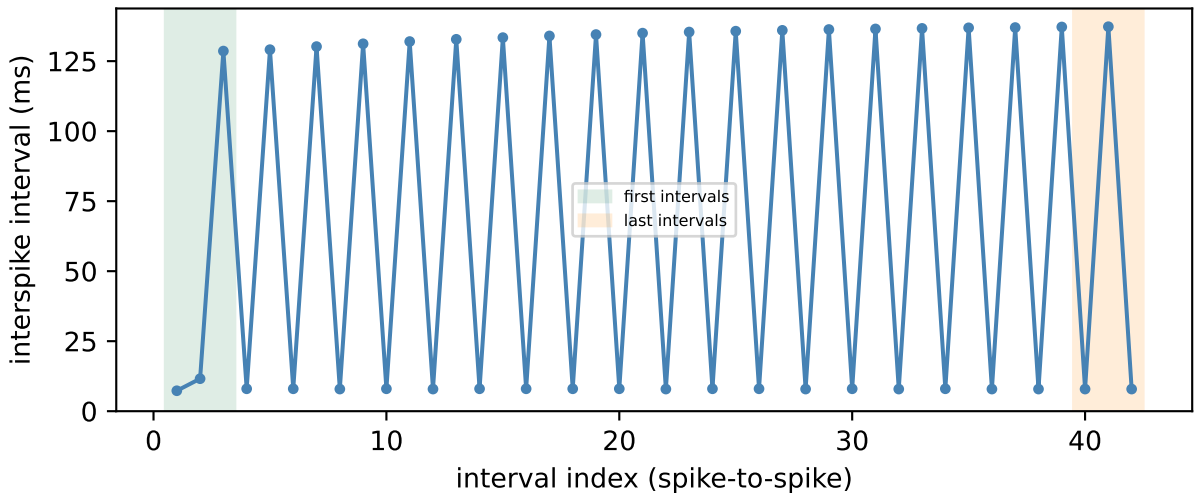
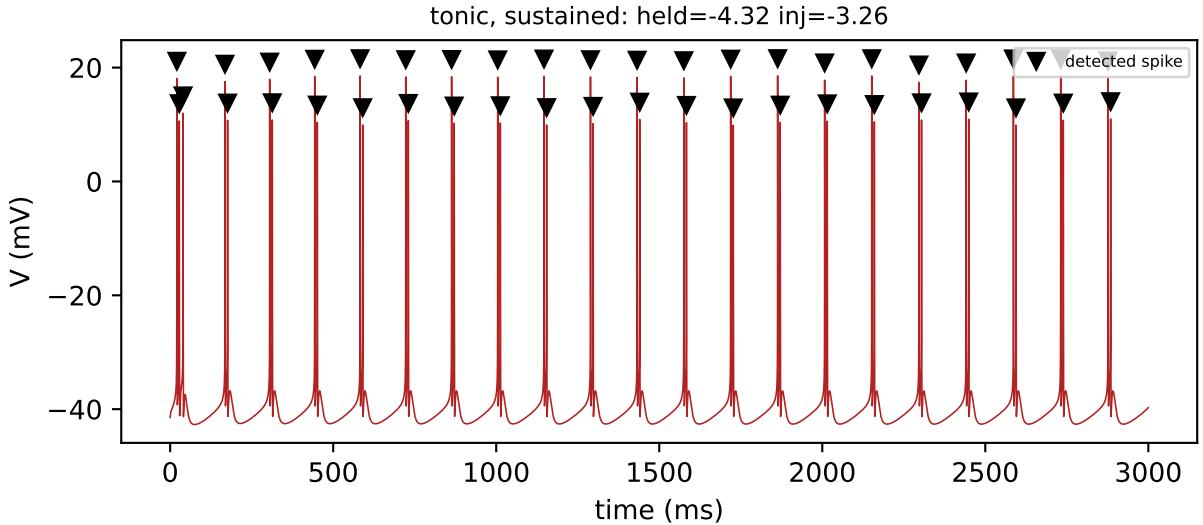
10. Sag depth

silent, deep sag: held=-3.54 inj=-5.50



Sag depth is the difference between the voltage just before the current step (-40.8 mV) and the most hyperpolarized point reached afterward (-43.8 mV), giving 3.0 mV of sag -- a hallmark of the hyperpolarization-activated current (I_h) partially repolarizing the cell during the step. The trough was found by searching the full 500 ms window (no spike occurred before it ended).

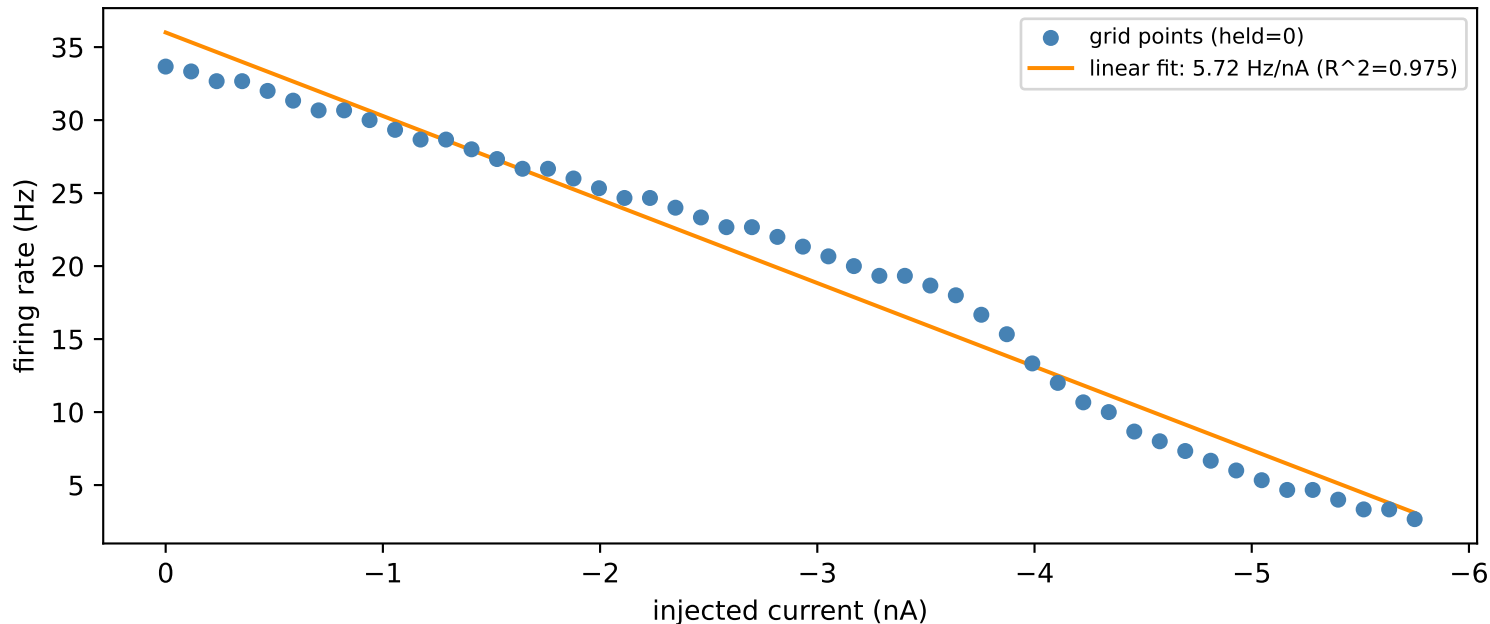
11. Spike-frequency adaptation



The adaptation ratio -- the mean of the last 3 interspike intervals divided by the mean of the first 3 -- is 1.04. Firing slows over the course of firing (spike-frequency adaptation) over this window.

12. Firing rate vs. injected current

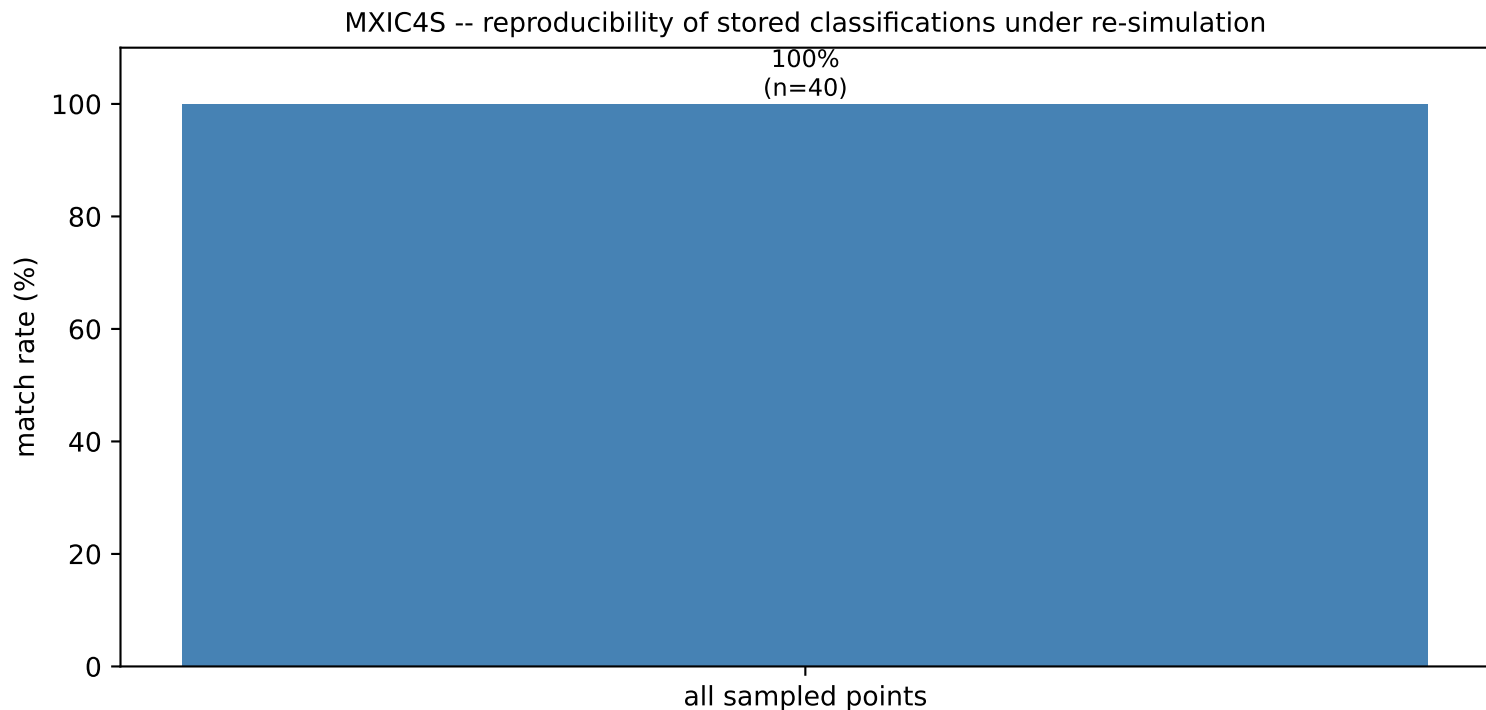
MXIC4S -- firing rate vs. injected current, no holding current



50 non-silent points with a clearly-classified firing pattern, sampled across injected current with no holding current applied (points still below firing threshold, which would otherwise bias the linear fit toward zero, are excluded). This firing-rate/current slope is one of the features used to characterize this cell in the cross-cell comparison; here it is plotted directly against the data it was fit to.

```
python src/generate_validation_packet.py --cell_id=MXIC4S --grid_dir=src/cell_data
source: cell_held_injected_grid.pkl -> cell_grid_features.pkl (extract_grid_features.compute_fit_slope())
```

13. Reproducibility check



40 random points from this cell's grid were re-simulated independently and compared against the classification stored from the original sweep. Match rate: 100%. Except at zero holding current, where the cell's own steady-state limit cycle is reused exactly, each holding-current level is reached by integrating forward from whichever already-settled level is numerically nearest -- this model is known to show path-dependent (hysteretic) settling near dynamical transitions, so a re-simulation that approaches a borderline point along a different path than the original sweep can occasionally land on a different classification right at that boundary.

`python src/generate_validation_packet.py --cells XB2IQX DFJ1K3 8Z3ESD MXIC4S`
source: cell_held_injected_grid.pkl -> cached test_pattern/rebound_pattern vs. fresh resimulate_point() results